

SECURE POLLING BOOTH USING RFID AND LIVE VOTE TRACKING

Abstract— As India is a country where the Democratic government rule is followed, every citizen has their right and duty to cast their votes. The existing methodology, Electronic Voting Machine EVMs was used for vote casting, which is not safe as data breaches and impersonisation of voting have been happening nowadays. To avoid these, our proposed methodology, “SECURE POLLING BOOTH USING RFID AND LIVE VOTE TRACKING” uses unique single-use RFID tags as a replacement for voter ID and two-factor authentication for casting the votes. The user scans their ID, which is linked with the database. After the verification process user will be requested to enter their 4-digit secret pin using buttons and displayed via the LCD screen. This acts as a two-factor authentication. To track the vote live, the Blynk IOT module is used, and the user will receive a SMS confirmation message after voting through the GSM module (Global System for Mobile Communication). To ensure single-time voting and secure polling.

I-INTRODUCTION

Each and every citizen has the right to cast their vote without any data breaches data threat and malware. Traditional voting systems such as Electronic voting machine [EVM] is used and it can't solve problems like vote tampering, proxy voting, data security and privacy issues. To overcome this, our proposed system “SECURE POLLING BOOTH USING RFID AND LIVE VOTE TRACKING” uses Arduino microcontroller for processing data, GSM module(Global System for Mobile communication) for sending confirmation message to the voter, an RFID reader is used to read the RFID card which is unique to each and every voter, 3X4 matrix keypad to enter their respective secret pin which ensures two factor authentication and push buttons to select their

candidates. System uses elder-friendly buttons to cast their votes, this helps senior citizens to cast their votes easily as many of them can't use the latest technologies. Blynk IoT platform is used to monitor and track the vote count lively resulting in accurate results.

II LITERATURE SURVEY

The usage of the proposed system significantly contributed to the improvement of existing electronic voting machine, initially it focuses on preventing vote tampering and live vote tracking.

Similarly, the paper curated by P.M. Benson Mansingh, T. Joby Titus V.S. Sanjana Devi titled “*A secured biometric voting system using RFID linked with the Aadhar database*” [1],[7],[9]demonstrates the usage and implementation of RFID cards linked with aadhar cards to showcase and protect the each voter in this process. It also reads the fingerprint of the voter to avoid vote tampering and proxy voting.

In addition the paper developed by Pradeep Kumara V.H., Ravindra P. Rajput titled Design and implementation of convenient and compulsory voting system using finger print sensor and GSM technologies [2],[5],[6],[10]finger print based authentication incorporated through GSM module is used here, Arduino was used for effective access and operation of the machine and the collected information will be transferred to nearby office for access through GSM module

Followed by the paper “On the Design and Implementation of a Blockchain Enabled E-Voting Application Within IoT-Oriented Smart Cities” [3] by Geethanjali Rathee, Razi Iqbal, Omer Waqar, Ali Kashif Bashir picturizes the blockchain and IoT in voting systems is maintained to prevent from future data modification by the smart devices. Also, to validate the proposed system various security

threats were prevented using access denial attack, service disruption and verification process.

Followed by the paper “Online voting system based on IoT and ethereum blockchain” [4] authored by Awsan A. H. Othman, Hamzah A. A. Muhammed, Emarn A. A. Muhammed, Prof. Mogeab A. A. Mosleh, Haneen K. M. Mujahid, picturizes the usage of blockchain and IoT in voting systems to ensure that user’s data is protected from data threat and eavesdropping. Also, the proposed system can be used for both government and private organizations by using it for conducting polls to real time voting polling booth.

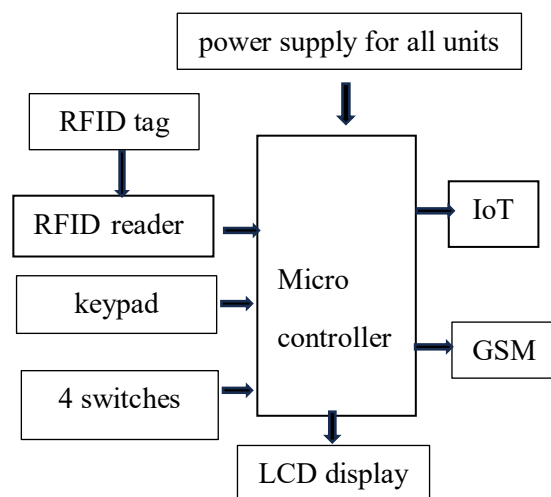
In the paper “Campus Choice: Arduino Nano College Voting Machine” [8] authored by Chaithanya D J, Sinchana S, Suprith N J, Guru kiran R. Features the fingerprint authentication with high degree accuracy and GSM module for ensuring transparency in the election. Also, an alert message will be sent to respective voter if his/her vote is misused by any intruder.

III-PROBLEM STATEMENT

The traditional voting systems such as ballot voting and machines like (EVM) Electronic Voting Machine may collect data of vote count but it can’t be ensured that the results are accurate and transparent. In the other hand there is no personal or unique identification method to address each voter ensuring the security and transparency during each voting. Since there is no unique identification to the voter there is a high risk of vote tampering, proxy voting and data breaches and security threats in the existing system. Also there is a high chance of voting more than one time which affects the transparency of voting system. It takes so much time to count all the votes as the authorized people are counting it manually which is not accurate resulting in wrong misleading information about the election results.

IV-PROPOSED MODEL

Our proposed system uses and ensures two factor authentication such as RFID scanning and entering a secret pin or password allowing the user to vote ensuring vote transparency and to prevent vote tampering. The instructions will be displayed via LCD screen so that the users can follow them. Arduino is used as microcontroller to process the information. Upon scanning the RFID card the voter is requested to enter their secret pin which acts as a verification process. Later the user will be instructed to press the vote button as each button represents a candidate to vote. IoT module is used to connect with Blynk software to track the vote lively. Also it helps in vote counting transparency and accuracy in vote results. GSM module is used send confirmation message to the voter. To avoid proxy voting RFID cards will be scanned only one time at the time of voting.



V-HARDWARE COMPONENTS

ARDUINO MICROCONTROLLER

Arduino ATMEGA328 works as a microcontroller in this system. It manages control, verification and communication of the system. Embedded Arduino board needs 16MHz with a 5V power supply with supported features like 32 KB flash memory& 1 KB EEPROM. For inputs, 6 analogue pins are used for memory, and for i/o pins, 14 digital pins are used. Arduino is accountable for validating the RFID card with the stored database and

The RFID(Radio Frequency Identification) reader and card, which acts as a voter ID and helps with the authentication process. Both the reader module and the card share a data transmission using radio waves. Each card will possess a unique ID number in its microchip, which makes each voter ID unique. After the card is read by the reader, the information will be sent to Arduino to compare with the database. If the UID matches the database, the voting process will be granted. Mitigating unauthorised access. The RFID cards are reusable and affordable, with a faster response, providing a tamper-proof voter verification experience

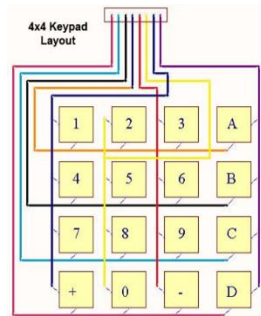
16x2 LCD display is perhaps the most used component to display text, which makes the project interactive. 16x2 refers to its ability to display a total of 32 characters simultaneously. It is based on liquid crystal display (LCD) technology and is normally operated with the



A keypad is an organised array of buttons in a matrix-like layout, which usually have numbers or letters on them and are pressed to give information to an electronic machine. It features a matrix-style keyboard. This arrangement reduces the number of microcontroller pins you need to connect. The most common is the 4x4 keypad of 16 keys (4 rows and 4 columns). The traditional telephone style 4x3 keypad featuring 12 keys is also popular in things like calculators and access control.

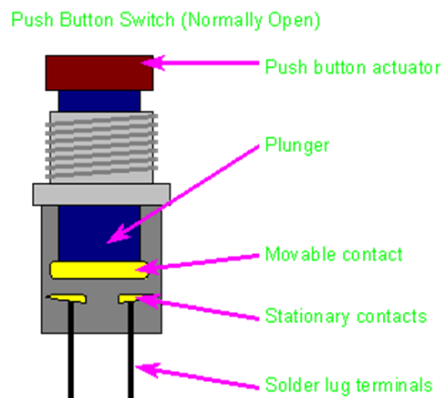
Upon triggering a key, it initiates the completion of the circuit between the selected row and column, helping the microcontroller to figure out which key was pressed by scanning the matrix. Keypads are inexpensive, durable and easy to connect to an Arduino, PIC, AVR or ARM MCU. They're all over the place in

ATMs, security systems, door locks, calculators, cell phones and other embedded applications where someone has to punch something. Although keypads do have speed limitations, and can only really process one or a few keys pressed at a time, they're still one of the most efficient and reliable input devices for small electronic applications.



PUSH BUTTON

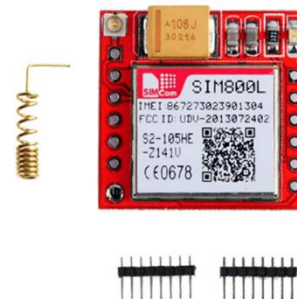
The push button plays a major role in selecting candidates for casting votes after the RFID card verification process. Each push button's action is monitored by the Arduino microcontroller for monitoring the vote count and updating it in the BLYNK software. Integration of push buttons helps in secured and tamper proof voting process



GSM Module (SIM800L)

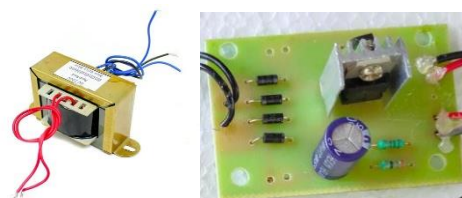
Global System for Mobile Communications (GSM) plays an unavoidable role in activating real-time communication between the voting system and the voter (user). The vote will be processed by Arduino and convert digital votes, into SMS to the voters with their registered mobile numbers. Through this GSM technology, the voters can receive the

confirmation SMS message, which acts as evidence ensuring one-time voting and transparency.



Power Supply (5V)

A 5V power supply plays an important role in providing a stable 5V current to the voting system, regardless of fluctuations in input. A 5V power supply is created and maintained using a step down transformer, rectifier and filter. The transformer reduces 230V AC voltage and converts it to DC voltage. A capacitor plays a major role in smoothing out ripples in the connection, thus helps the power supply to distribute the current equally.



IoT (ESP8266)

The ESP8266 IoT module has a built-in microcontroller supported and powered using the Arduino IDE module with a 3.3V power and GPIO pins, which helps in the interfacing process with sensors. The acts as a bridge for communication between the physical devices and application platforms, helping in enabling live vote tracking. The IoT module completes the task of voting and monitoring the vote, also used for data storage and access purposes.



VI-RESULTS

Our system's secure polling booth using RFID and live vote tracking performed well during testing. The voters received a message after casting their vote successfully, which will be considered a confirmation message. The cards are read only once to avoid vote tampering and proxy voting, and the transparency of the vote count is maintained using the Blynk IoT platform so that the election officials can track the vote count live. Instructions are given to the voters to complete the step-by-step procedure via LCD screen to make the voting experience easier.

VII-FUTURE ENHANCEMENT

In the future, this project can be further enhanced and expanded to improve its scalability, compatibility and consistency. Some possible advancements include:

1. Blockchain Integration

Blockchain integration helps in improving the security, reliability and consistency of the voting system.

2. Enhanced Cloud Dashboard.

A private cloud enhanced dashboard can be used for vote monitoring and data security purposes.

3. Offline Data Backup

Offline data backup can be done using SD cards and local EEPROM storage. This practice helps in preventing losing vote counts, whenever a poor internet connection happens.

4. Solar-Powered Operation

Solar panels and rechargeable batteries can be used to enhance sustainability to work without the help of electricity.

5. Multilingual and Voice-Assisted Interface

A multilingual and voice-assisted interface will make it more user-friendly, as senior citizens who are unaware of technologies can cast their vote easily.

These methodologies can be added as future enhancements to make the voting process reliable, fast and more secure.

VIII-CONCLUSION

The proposed system, “Secure pooling system using RFID and live vote tracking”, demonstrates and illustrates the importance of preventing vote tampering and proxy voting during elections. By combining technologies, a GSM module for sending confirmation messages, an IoT module for vote tracking made voting as transparent and tamper-proof.

Through these technologies, the system gains accuracy, security, transparency and compatibility in the voting process and reduces manual effort for vote counting and human error. It provides a digital transformation of the voting experience for both voters and administrators. Overall, this project proves to be a step toward secured and transparent electronic voting systems, suitable from small to large scale industries with further optimization and development of this project in future.

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